RKDF University, Bhopal
Faculty Profile

Basic Information	
Name	**Dr. VINOD KRISHNA SETHI**
Date of Birth	**21 July 1949**
Designation	DG (research)
Department	R&D Cell
Experience	Power Industry (Govt. of India) …30 years Teaching & Research (Academics) :….24 Years Total ,,, 54 Years
Email ID	vksethi1949@gmail.com
Contact No	9713902378

Educational Qualifications (refer Annexure I)			
Description	**Year**	**%**	**Institute/University**
(UG)- B Sc (Hons)- Maths., Phy, Statistics **B.E (Mechanical)**	**1967** **1971**	**79%** **82%**	**Agra University** **University of Roorkee** **Now IIT (Roorkee)**
(PG)-Course in Power Engg	**1978**		**Salford College ,** **UMIST, Manchester UK**
M. Phil.			
(Ph. D.)-	**1986**		**IIT Delhi**
Post Doctorate			
NET Qualified/GATE			

Experience Detail (Refer Annexure I)				
Experience (Teaching/Research)	**Designation**	**Duration**		**Name of Institute/University**
		From	**To**	
3 and half years	**Research Advisor**	**Dec 1999**	**July 2003**	**Osmania University Hyderabad**
11 tears	**Director**	**July 2003**	**July 2014**	**RGPV (State Technological University of MP)**

4 and half years	**Vice Chancellor**	**Jan 2015**	**July 2019**	**RKDF University, Bhopal**
5 years	**DG Research**	**July 2019**	**Till date**	**RKDF University , Bhopal**

Publications

No. of Papers Published (Attach Annexure of papers published in peer-reviewed or refereed journals)	**Total 163 papers Annexure II attached**
No. of Books Published	15
Books Chapters Published (Provide print of 1st Page of Book	01
No. of Patents Published/Grant	5

Ph. D/M. Phil Project supervised

Research Program	**Award**	**Under Supervision**	**Name of University**
Ph. D (Provide detail i.e. name, title etc)	30	NIL	RGPV University…25 RKDF University …5
M. Phil			
PG Thesis/Dissertation			

Area of Expertise (100 words)

- **I am Retired IES (UPSC 1974 Batch) and Power Engineer having total about 30 years of experience (Energy & P0wer… Nuclear (BARC) 4 years, Power Ministry (CEA)…26 years)**
- **I have been in key positions of Govt. of India as Director / Chief Consultant at Power Projects at Delhi (Badarpur) and Singrauli Region (ANPARA)**
- **In the field of Academics I have been at professor level for 24 years as Director , Vice Chancellor & DGR**
- **Presently I am Chair Professor at Sir J C Bose Chair & Member of Monitoring Committee of CSIR Govt. of India**

Award and Achievement

Name of Award	**Description (With certified of Copy award)**
National	**Jewel of India, Life Time Achievers Award & Mother Theresa Award (IBS)**

International	NIL	
Conference/Seminar/Workshops/FDP		
Description	**No.**	
Conference/Seminar paper presentation	**46**	
Conference/Seminar attended/ organized	74	
Work shop attended/ organized	23	
FDP Attended/ organized	35	
Research Project (Refer annexure I)		
Name of Project	**Funding Agencies**	**Amount(Rs. In LAKH)**
Power Productivity Enhancement through Hybrid of Solar Wind Hybrid	AICTE	3.00
Simulation & optimization of surface roughness ir Gas turbine cooling channel Computational Fluid Dynamics (CFD)	AICTE	19.50
Modeling& Simulation of Carbon Recycling Technology through Conversion of CO2 in flue Gas into Useful Multipurpose fuel	DST	25.83
Design & Development of Nano Technology based Solar PV cells	MPCST , BHOPAL	8.00
60 kW Solar wind hybrid System at RGTU Hill	MNRE	100.3
Process Stabilization, Evaluation and Analysis of CO2, capture and its conversion into fuel molecules CO, H2, CH4 using pilot plant of CO2 capture and sequestration at RGPV	DST	15.00
30 kW CL-CSP Solar Thermal Plant	MNRE	976
High Energy Density Thermal Energy Storage for concentrated Solar Plant	MNRE	41.00
Post Combustion Carbon Capture & sequestration (CCS) Plant on a Coal Fired Thermal Power Plant – Feasibility Study	CPRI Ministry of Power, Go	53.5
System Design, Erection, Testing & Commissioning of 40 kWth and 10 kWe pilot plant aiming at the Feasibility Study of MWe Scale Concentrated Solar Thermal Plant integrated with 24 x 7 Thermal Energy Storage.	MNRE	81.50

- **Any other Achievement**
  Presently I am Member of Monitoring Committee of CSIR Govt. of India in CCUS Mission (HCP-48) & I am Founder PI of India's first Carbon Capture Plant

Signature

ANNEXURE -II

LIST OF PUBLICATIONS IN GREEN POWER

(Dr Vinod Krishna Sethi)

I. INTERNATIONAL JOURNALS

1. **Sethi, V.K., & Sharma, P.B. (1985). A model for combustion losses in a pulverized fuel fired power plant boiler. *Proc. I.Mech.E. (London), 202*(A4').**
2. **Sethi, V.K., & Sharma, P.B. (1990, U.K.). A study of transients in turbine heat rate of steam power plant. *Energy Conversion & Management, Boston, Texas. , 2,* 187-202.**
3. **Sethi, V.K., & Sharma, P.B. (1984). Computer aided optimization of turbine heat rate of a thermal power plant. *Trans. ASME. Jr. of Energy Resources*.**
4. **Sethi, V.K., & Sharma, P.B. (1995). Improved formulation for turbine heat rate of a thermal power plant. *'Energy' International Journal*.**
5. **Reddy, D.N., &Sethi, V.K. (2002). Renovation and refurbishment of old polluting power plants using CFBC Boilers – Technology status in India. *Journal of 8th European Roundtable on Cleaner Production, Cork, Ireland*.**
6. **Reddy, D.N., &Sethi, V.K. (2006). Green power technologies & CDM opportunities – A case study. *SESI Journal*.**
7. **Sethi, V. K., et al. (2007). Power productivity enhancement & statistical analysis is using hybrid system of solar wind & biomass. *Asian Journal of Experimental Sciences*.**
8. **Sethi, V. K., et al. (2007). Production potential of electricity from biomass in Indian context. *Ultra Science, 19*(1), 1-10.**
9. **Sethi, V.K., et al. (2008). Wind resource assessment and site analysis at RGPV, Bhopal. International conference on environmental research (ICER-08) Dec. 18-20 (2008), International Journal of Environmental Research &Development (JERAD).**
10. **Sethi, V.K., et al. (2009). Osmotic stress in the CynobacteriumNosticMuscorum overcomes by the accumulation of proline. *Canadian Jl. of applied science, 3*(1).**
11. **Sethi, V.K., et al. (2009).Extrusion process design for non-circular rods. *International Journal of Applied Engineering Research ISSN 0973-4562,4*(10), 2063-2070.**
12. **Sethi, V.K., et al. (2009).Process and die design for rod extrusion of γ iron. *International J. Matter form,2*(3), 191-196.doi :10.1007/s12289-009-0403-2.**
13. **Sethi, V.K., et al. (2010). Analysis of wind power potential on complex terrain by flow modelling and times series characteristics. *International Journal of Engineering Research and Industrial Applications (IJERIA,) 3*(I), 115-130.**
14. **Sethi, V.K., Gupta, B., et al. (2012). Effect of air velocity fuel rate and moisture content on the performance of updraft biomass gasifier using fluent tool. *International Journal of Modern Engineering Research (IJMER), 2*(5), 3622-3627.**
15. **Sethi, V.K., et.al. (2009). Mathematical simulation and energy estimation of 10 kW horizontal axis wind turbine rotor at hilly site of RGPV, Bhopal (Case Study). *Journal of Current World Environment, 4*(2), 255-262.**
16. **Sethi, V.K., et al. (2010).A novel approach for CO2 sequestration and conversion in to useful multipurpose fuel. *International Journal ICER-JERAD*, 5(3A), 732-736.**
17. **Sethi, V.K., Shukla, P., et al. (2010).CNT enhances the efficiency of solar PV cell. *Search & Research interdisciplinary journal, 1*(3), 6-9.**
18. **Sethi, V.K., Shukla P., et al. (2010). Silicon nano-particle enhance performance of solar PV cell. *Search & Research, 2*(1), 91-93.**
19. **Sethi, V.K., Shukla P., et al. (2016). Evaluation of thin film amorphous silicon solar energy system. *Industrial Engineering letters, 6*(4).**
20. **Sethi, V.K., Shukla P., et al. (2013).Modelling of thin film a-Si solar energy system. *Journal of Sustain Manufacturing and Renewable Energy, 2*(3-4), 121-135.**
21. **Sethi, V.K., Shukla, P., et al. (2011). Concepts of efficiency enhancement in silicon solar cell. *Journal of Sustainable Manufacturing and Renewable Energy, 1*(1-2).**
22. **Sethi, V.K., Shukla, P., et al. (2011). Use of nanotechnology in solar PV cell. *International Journal of Chemical Engineering and Applications, 2*(2), 77-80.**
23. **Sethi, V.K., Shukla, P., et al. (2011). Cost boundary in silicon solar panel. *International Journal of Chemical Engineering and Applications, 2*(5), 372-375.**

24 **Sethi, V.K., Shukla P., et al. (2012). Thin film photo-voltaic cell compared to mono crystalline photovoltaic cell and multi crystalline photo-voltaic cell. *International Journal of Advanced Renewable Energy Research, 1*(2), 26-35.**
25 **Sethi, V.K., Pandey, M., Gupta, B., et al. (2012). Parametric study of fixed bed biomass gasifier: A review. *International Journal of Current Engineering &Technology, 2*(1).**
26 **Sethi, V.K., Pandey, M., Gupta, B., etal. (2012). Polymer exchange membrane (PEM) fuel cell: A review. *International Journal of Thermal Technology, 2*(1).**
27 **Sethi, V.K., Shukla, P., et al. (2013). Lab view based modeling of amorphous silicon solar panel. *TIT International Journal of Science & Technology, 2*(2).**
28 **Sethi, V.K., Kurchania, R.,& Khan, J.M. (2010).Synthesis and characterization of two dimensional grapheme lamellae based pan nano-composites. *Elsevier, 519*(3), 1059-1065.**
29 **Sethi, V.K., Shukla, P., et al. (2012). Concentrating solar power, seawater desalination, parabolic troughs, fresnel systems. *International Journal of Advanced Renewable Energy Research, 1*(3),25-31.**
30 **Sethi, V.K., Shukla, P., et al. (2012). Recent trends in dye-sensitized solar cell technology. *International Journal of Sustainable Manufacturing and Renewable Energy, 1*(3-4), 121-129.**
31 **Sethi, V.K., Shukla, P., et al. (2012). Development and performance, evaluation of a-Si thin-film PV cell. *Nano materials and Energy, 1*(6), 338-344.**
32 **Sethi, V.K., Shukla, P., et al. (2012).Progress in thin film silicon based solar cell technology.*International Journal of Advanced Renewable Energy Research, 1*(7), 429-433.**
33 **Sethi, V.K., Jain, P., et al. (2011). Optimization of the performance of the wind power generation unit by using different neural networks. *International Journal of Power System Operation and Energy Management, 2*(3, 4), *2231-4407*.**
34 **Sethi, V.K., Thapar, V., et al. (2011). Critical analysis of methods for mathematical modeling of wind turbines. *Elsevier - ISSN Renewable Energy, 36*(11), 3166-3177.**
35 **Sethi, V.K., Thapar, V., et al. (2011). Estimation of hourly temperature at a site and its impact on energy yield of a PV module. *International Journal of Green Energy, 9*(6), 553-572.**
36 **Sethi, V.K., et al. (2010). Silicon nano particle enhance performance of solar PV cell. *Search & Research, 2*(1), 91-93.**
37 **Sethi, V.K., Pandey, M., Jain, P., et al. (2013). Technology analysis of concentrated Solar thermal power. *Innovative System Design Engineering IISTE, 4*(6), 75-80.**
38 **Sethi, V.K., et al. (2013). An investigation of wind characteristics for optimum wind energy utilization at the campus of the Rajiv Gandhi Technical University, Bhopal (India). *International Journal of Engineering Sciences & Research Technology (IJESRT), ISSN: 2277-9655*, 3067-3072.**
39 **Sethi, V.K., et al. (2013). An assessment of wind energy potential for design and implementation of a wind power project. *International Journal of Emerging Technology and advanced Engg. (IJETAE), 3*(10), 569-574.**
40 **Sethi, V.K., et al. (2014). Design optimization of solar PV power plant for improved efficiency of solar PV plant by maximum power point tracking system. *International Journal of Research in Engineering and Technology (IJERT), 3*(1), ISSN: 2278-0181.**
41 **Sethi, V.K., et al. (2013). Modeling of thin film a-Si solar energy system. *Journal of Sustain Manufacturing and Renewable Energy, 2(3-4,)*ISSN: 2153-6821.**
42 **Sethi, V.K., et al. (2012). Development and performance, evaluation of a-Si thin-film PV cell. *Nano-materials and Energy, 1* (NME6), 338-344.**
43 **Sethi, V.K., et al. (2012). Thin-Film photovoltaic cell compared to mono crystalline photovoltaic cell and multi crystalline photovoltaic cell. *International Journal of Advanced Renewable Energy Research, 1*(2), 26-35.**
44 **Sahay, A., Sethi, V.K., Tiwari, A.C. (2013). A comparative study of attributes of thin film and crystalline photovoltaic cells. *VSRD International Journal of Mechanical, Civil, Automobile & Production Engineering, 3(7),* 267-270.**
45 **Sahay, A., Sethi, V.K., Tiwari A.C. (2013). Cost performance of thin film and crystalline photovoltaic cells - A comparative study. *International Journal of Current Engineering and Technology, ISSN 2277-4106, 3*(3), 1139-1142.**
46 **Sahay, A., Sethi, V.K., Tiwari, A.C. (2013). Design, optimisation, and system integration of low cost Ground Coupled Central Panel Cooling System (GC-CPCS). *International Journal of Current Engineering and Technology, ISSN 2277-4106, 3*(4), 1473-1479.**

47 **Sahay, A., Sethi, V.K., Tiwari, A.C. (2014). Fabrication scheme, instrumentation scheme and testing of Ground Coupled Central Panel Cooling System (GC-CPCS).** ***International Journal of Current Engineering and Technology, ISSN 2277-4106, 4*****(2), 631-638.**

48 **Sethi, V.K., et al. (2012). A general design procedure for ground coupled heat exchanger (GCHEX) and its application to ground coupled central panel cooling system (GC-CPCS).** ***Renewable and Sustainable Energy Reviews, ISSN 1364-0321, 42*****, 306-312.**

49 **Vyas, S., Sethi, V.K., &Chouhan, J.S. (2016). Process flow and analysis of CCS plant installed at RGPV Bhopal run by biomass gasifier.** ***International Journal of Mechanical Engineering and Technology (IJMET), 7*****(3), 387–395.**

50 **Sethi, V.K., Vyas, S. (2017). An innovative approach for Carbon capture & Sequestration on a thermal power plant through conversion to multi-purpose fuels – A feasibility study in Indian context.** ***Science Direct, Energy Procedia, 114*****, 1288-1296.**

51 **Sethi, V.K., et al. (2016). Baseline creation for carbon dioxide capture and sequestration plant using Mono-ethanolamine Absorbent.** ***International Journal of Current Engineering and Technology, E-ISSN 2277 – 4106, P-ISSN 2347 – 5161, 6*****(3).**

52 **Vyas, S., Sethi, V.K. (2017). Performance evaluation of MEA based CCS Plant installed at RGPV.** ***Jr. of ECDS - IEEE,*** **(4), 348-352.**

53 **Sahay, A., Sethi, V.K., Tiwari, A.C. & Pandey, M. (2017). Relevance and cost benefit analysis of Ground Coupled Central Panel Cooling System (GC - CPCS).** ***J. Inst. Eng. India (Springer) Ser. C*****, 1-6, ISSN No. 2250-0545, DOI: 10.1007/s40032-017-0362-1.**

54 **Sahay, A., Sethi, V.K., Tiwari, A.C. & Pandey, M. (2015). A review of solar photovoltaic panel cooling system with special reference to Ground coupled central cooling system (GC - CPCS).** ***ELSEVIER, Renewable and Sustainable Energy Reviews 42,*** **306 – 312.**

55 **Bajpai, S., & Sethi, V.K. (2017). An overview of solar roof top program in India with specific reference of Madhya Pradesh policy for decentralized renewable energy.** ***IJRASET, 5*****(VIII), 2315- 2320, ISSN: 2321- 9653.**

56 **Verma, S.K., Lodhi, G.S., & Sethi, V.K. (2017). Strategic Planning of Solar Parks in Madhya Pradesh, India.** ***International Journal of Research and Scientific Innovation, (IJRSI), IV*****(VIII), 24-31, ISSN 2321-2705.**

57 **Sahay, A., Sethi, V.K., Tiwary, A.C., & Pandey, M. (2017). Commercial viability of cooling of solar photovoltaic panels using Ground Coupled Central Panel Cooling System (GC - CPCS).*****International Jr. of Current Engineering and Technology (IJCET)*****, 7(05), 1885**

58 **SurendraBajpai, Dr.V.K.Sethi(2017). Review: Design, Simulation and Economic Analysis for Decentralized and Distributed Power Generation in India.** ***International journal on Emerging Technologies*** **8 (1):70-78 (2017).**

61 **RupanshuSuhane. Dr. V K Sethi et. al. (2018). Mathematical Modeling of Battery and Ultra Capacitor for Photo Voltaic System,*****International Journal of Engineering & Technology,7 (3.27) (2018) 41-46.***

62 **Ruchi Pandey 1, V K Sethi 2, Mukesh Pandey 3, ManeeshChoubey4 , "Recent Research and various Techniques available for Efficiency Improvement of IGCC Power Plants" published in "Springer AISC Series/SCOUPS INDEXED JOURNAL "Artificial Intelligence and Evolutionary Computations in Engineering Systems" pp 49-58 April 2015.**

63 **Ruchi Pandey 1, V K Sethi 2, Mukesh Pandey 3, ManeeshChoubey4 , "Comparative Study of Coal Based IGCC and Conventional Power Plants on various Technological Parameters" published in IEEE Journal of "Advances in Computing and Communication Engineering" (ICACCE 2015)" in May 2015**

64 **Ruchi Pandey 1, V K Sethi 2, Mukesh Pandey 3, ManeeshChoubey4 ,"Modelling and Simulation of Integrated Gasification Combined Cycle (IGCC) Power Plant and its Efficiency Optimization" published in "Materials Today" Elsevier, March 2016, pages 1-5**

65 **Ruchi Pandey 1, Mukesh Pandey 2 ,V K Sethi 3, ManeeshChoubey4 "A 100 MW Integrated Gasification Combined Cycle (IGCC) Power Plant: Efficiency Optimizationpublished in Elsevier Journal of Advanced Research in Dynamical & Control Systems, Volume 9 Issue 6, 2017 Pages 212 - 220**

66 **Ruchi Pandey 1, Mukesh Pandey 2, V K Sethi 3, ManeeshChoubey4 , "Technology and**

Resource Management of IGCC Power Plants in the Indian Context" published in Elsevier Journal of Advanced Research in Dynamical & Control Systems, Vol. 9, No. 3, 2017, Pages: **108-120**

67 Ruchi Pandey 1, Mukesh Pandey 2 ,V K Sethi 3, ManeeshChoubey4 "Potential Analysis of Indian Coal for Power Generation and Pollution Based on Integrated Gasification Combined Cycle", published in Elsevier Journal of Advanced Research in Dynamical & Control Systems, Volume 9 Issue 6, 2017, Pages 212 - 220.

II. INTERNATIONAL CONFERENCE PAPERS

1. **Reddy D. N. &Sethi V. K., "IGCC for Power generation – an Environmentally benign and Energy efficient Technology" ICERD 2, Kuwait, April 8-10, 2002.**
2. **Reddy D.N., Sethi V.K. "Fluidized bed technologies to High Ash Indian Coals – A Techno-economic Evaluation", 8th European Roundtable on Cleaner Production, Cork, Ireland Oct 9-11, 2002.**
3. **Reddy D.N., Basu K and Sethi V.K. "IGCC for Power Generation- A promising Technology for India". International Conference on Clean Coal Technologies for Our Future, Sardinia, Italy, 21-23 Oct. 2003.**
4. **Sethi V.K. et. al. "Technology & Resource Management of IGCC Power Plant in the Indian Context- A Case Study" -Twenty first Annual International Pittsburgh Coal Conference, Osaka Japan, Sept. 13-17, 2004.**
5. **Reddy D.N., Sethi V.K., Rajesh P. "Technology options for Indian Coals and Optimization of IGCC Cycle efficiency" -Twenty first Annual International Pittsburgh Coal Conference, Osaka Japan, Sept. 13-17, 2004.**
6. **Sethi V.K. "A Road map to Sustainable Power Development in India over Next Five Years, GLOGIFT-05, International Conference RGTU, Bhopal, Dec.2005".**
7. **Baredar Prashant, Sethi V.K., Pandey Mukesh, 'Wind resource assessment and site analysis power Generation through hybrid system of solar, wind & Biomass' 5thGlobal international con. on Flexible systems Management organized by GLOGIFT & RGPV**
8. **Sethi V.K., ShrivastavaSanjeev and Baredar P. "Green Power generation through Hybrid System of Solar, Wind & Biomass." GLOGIFT-05, International Conference, RGTU, Bhopal, Dec- 2005.**
9. **Sethi V.K. et. al. "Solar – Wind Hybrid System: A viable option for Energy Security and source of Carbon Credits", International Conference on Wind Energy: Trends and Issues 5-7 January 2006, NITTR, Bhopal.**
10. **Sethi V.K. et. al. "CDM opportunities in Renewable Energy Projects with particular reference to Wind Power", International Conference on Wind Energy: Trends and Issues NITTR, Bhopal, 5-7 January 2006.**
11. **Reddy D.N., Sethi V. K. "Green Power Technologies for high Ash Indian Coals Evaluation of CO_2 mitigation options" 23rd International Pittsburgh Coal conference, Sept. 25-28, 2006, USA.**
12. **Reddy D.N., Sethi V. K. "Refurbishment of old PC Boilers confronted with Coal Quality Degradation by a CFBC boiler" 23rd International Pittsburgh Coal conference, Sept. 25-28, 2006 USA.**
13. **Sethi V. K. et. al. "Turbine Heat Rate under load transients of Al-zara Thermal Power Station, Syria" International Conference on Modeling and simulation, Coimbatore, 2007.**
14. **Sethi V. K. et. al. "Mathematical Modeling of Turbine Heat Rate under steady state conditions" International Conference on Modeling and simulation, Coimbatore, 2007.**
15. **Sethi V. K. et. al. "Effects of Load, Leak off Steam consumption and Main Steam properties on Turbine Heat Rate for Al-Zara Thermal Power Station, Syria" International Conference on Modeling and simulation, Coimbatore, 2007.**
16. **Baredar Prashant, Sethi V.K., Pandey Mukesh, 'Power Productivity Enhancement and Reliability Analysis Using Hybrid System of Solar, Wind and Biomass International conference on environmental research (ICER-07) 28-30 Dec. 2007 At Govt. Geetanjali Girls P.G.College Bhopal**

17. **Baredar Prashant, Sethi V.K., Pandey Mukesh 'Overview on worldwide developments in wind power Technology' 6thGlobal International conference on Flexible systems Management organized by GLOGIFT & UPTU 15-17 Nov. 2007 at Noida(U.P.)**
18. **Sethi V. K. et. al. "Situation and SWOT Analysis of Small Wind- Solar- Biomass Tribid Energy System Installed At University Institute of Technology, Bhopal M.P., (India)" Proceedings of GLOGIFT 08 June 14-16, 2008 at Stevens Institute of Technology Hoboken, NJ, USA pp. 1-9**
19. **Sharma P B &Sethi V. K. "Mitigating Climate Change through Green Energy Technologies" International seminar on New Horizons of Mechanical Engineering (ISME 2008), March 18-20, 2008 at Rajiv Gandhi Technological University, Bhopal, M.P.**
20. **Sethi V K et. Al. "Modeling& Simulation of Carbon Recycling Technology through Conversion of CO_2 into useful multipurpose fuel" International seminar on New Horizons of Mechanical Engineering (ISME 2008), March 18-20, 2008 at Rajiv Gandhi Technological University, Bhopal, M.P.**
21. **Sethi V.K., Pandey Mukesh, Priti Shukla "Use of Quantum Dots in Solar Panel" 15th ISME International Conference on Advances in Mechanical Engineering organized by Rajiv Gandhi Technical University on 18th - 20th March 2008**
22. **Baredar P., Sethi V.K., Pandey Mukesh 'Statistical Analysis of Solar-Wind Hybrid System Using Systat Software', International online Energy Conference (ioec-08) 19 Dec 2008 at the University of Leeds, UK**
23. **Baredar Prashant, Sethi V.K., Pandey Mukesh Wind resource assessment and site Analysis at Rajiv Gandhi Technological University, Bhopal, International conference on Environmental Research (ICER-08) Dec. 18-20 (2008) At Goa.**
24. **V K Sethi et al. "Use of CNT & Solar PV Cell" Proceedings of International Conference in Advances & Renewable Energy, ICARE 2010**
25. **ManjuKhare, Yogendra Kumar, Ganga Agnihotri, V.K. Sethi "Modeling of Photovoltaic Module/Array Using MATLAB/SIMULINK". Proc. of Int. Conf. on "Trends & Advances in Computation & engineering" (TRACE), Feb 2010,pp.183-188.**

26. **ManjuKhare, Yogendra Kumar, Ganga Agnihotri, V.K. Sethi "Energy Storage. Technologies for Integrated Renewable Energy Sources - A Technical Review" in Proc. of Int. Conf. on "Electrical power and Energy systems" (ICEPES – 2010), August 26-28, 2010**

27. **Sethi V.K., Pandey Mukesh, Priti Shukla "Concepts of High Efficiency Silicon Solar Cell" International Conference on concurrent Techno &Envirosearch"Search And Research Youth Congress-2010, 4th-5th Dec.2010**

28. **Vyas S, Sethi VK, Chouhan JS, Sood A. "Prospects of Integrated Collaborative Technology Of Carbon Dioxide Capture", 32nd National Convention of Environmental Engineers: Challenges in Environment Management of Growing Urbanization, IEI 2016. doi:10.13140/RG.2.1.3606.8086.**
29. **Anil Kumar, V.K. Sethi. (2018). Solar Thermal Plants coupled with a Thermal Storage- a Review of Thermal Storage options.** ***International Conference on Energy, Environment and Economics, 14-16 August 2018***
30. **Dr.Vinod Krishna Sethi, et al. A Pilot Study of Post Combustion Carbon Capture & Sequestration Pilot Plant aimed at Feasibility Analysis of Installation of a Carbon Capture, Utilization and Sequestration (CCUS) Plant on a Large Thermal Unit in India.** ***14th International Conference on Greenhouse Gas Control Technologies, GHGT- 14 21st -25th October, 2018 Melbourne, Australia.***

III. NATIONAL CONFERENCES/ PROCEEDINGS / JOURNALS

1. **Dr. V. K. Sethiet. el. , "CO2 Mitigation and CDM Through Green Power Technologies" National Conference on Clean development Mechanism, 10-11 Feb. 2005**
2. **Dr. V. K. Sethiet. el. "CDM Opportunities in Municipal Solid Waste to Energy Projects" National Conference on Clean development Mechanism, 10-11 Feb. 2005**

3. **Sethi V.K. et. el. , "Transfer of Technology of A Green Power Project From a Developed Economy to India", Conf. on Emerging Opportunities for Consulting Service Providers… January 12-13, 2005**
4. **Sethi V.K. et. el., Green Power Engineering for Sustainable Power development in India, Proc. of INMANTEC, Delhi**
5. **Sharma P B, Sethi V K, "Power Plant Performance Analysis & Optimization Through Mathematical Modeling", All India Technical Program n Performance Operational Analysis and Diagnosis and Optimization of Thermal Power Plant Equipments, 24-25 November 2006, Hyderabad.**
6. **Sethi V K, Sharma P B, Reddy D N, "A Transition from Conventional Thermal to Green Mega Power", All India Technical Program n Performance Operational Analysis and Diagnosis and Optimization of Thermal Power Plant Equipments, 24-25 November 2006, Hyderabad.**
7. **Sethi V K, et. al., "Commercialization of Biodiesel", National Seminar on Rural Technology, 2006, Bhopal.**
8. **Sharma P. B. &Sethi V. K. "Mathematical Modeling of Turbine Heat Rate under steady state conditions" CFD the New third dimension in flow analysis and thermal design, RGTU, Bhopal, 2007.**
9. **Sharma P.B. &Sethi V. K. "A Transition from Conventional Thermal to Green Mega Power", GE Seminar, DREAMS 2007.**
10. **Baredar Prashant, Sethi V.K., Pandey Mukesh "Feasibility Analysis 0f Pilot Hybrid System Installed at RGTU Bhopal", All India Young scientist conference Nov.23-25, 2007 at P.V.Naronha Administration Academy, Bhopal, M.P.**
11. **Sethi V.K., Pandey Mukesh, Priti Shukla "The use of Hydrogen Energy in Vehicles" National Seminar on Bhartiya VigyanSammelan 2007, Bhopal**
12. **Sethi V.K., Pandey Mukesh, Priti Shukla "Silicon Nano-particles enhance performance of Solar cell" presented in National Seminar on Efficient Conversion of New and Renewable Energy for Electricity & Fuel on 22nd – 23rd Dec.2008.**
13. **Sethi VK et el. " CNT Enhances the efficiency of solar PV cell" presented in National Seminar on 2nd BhartiyaVigyanSammelan 2009 organized by VijnanaBharti, M.P.C.S.T., Govt. of M.P. on 1-3, Dec 2009**
14. **V K Sethi etal. "Design and Modeling of Wind Farm and Predicting Annual Energy Production in a Hilly Terrain of RGPV, Bhopal, Madhya Pradesh" MNIT Conf. March 2010**
15. **ManjuKhare ,Yogendra Kumar , "A Study of Maximum Power Point Tracking Controllers in Photovoltaic Application" in Proc. of National Conf. on "Advance Trends in Electrical Engg.,26-27 march2010, pp.53-59.**
16. **V K Sethi et al. "Concepts Of High Efficiency Silicon Solar Cell" presented in International Conference onConcurrent Techno & Environ search "Search And Research Youth Congress-2010 "Organized by Search And Research Group& Search And Research Journal ,Bhopal on 4th-5th Dec.2010**
17. **V K Sethi et al. "Use of CNT in Solar PV cell" presented in International Conference on "advances in Renewable Energy" (ICARE-2010) organized by MANIT, Bhopal on 24th -26th June2010.**
18. **V K Sethi et al. " Flexibility in Service Sector" presentedin Ninth Global Conference (GLOGIFT 09) On Flexible System Management organized by NITIE , Mumbai on 12th – 14th Nov.2009.**
19. **V K Sethi et al. "Use of Quantum Dots in Solar Panel" presented in 15th ISME International Conference on Advances in Mechanical Engineering organized by RGPV**
20. **Sanjay Kumar Verma.V K Sethi et al. (2018). Planning of Rewa Ultra Mega Solar Power Project (750MW) at Rewa, Madhya Pradesh.** ***Energy Next Magazine(s), May 2017, April 2017, June 2017); (MNRE Scheme/ Documents)***

PUBLICATIONS IN THE AREA OF THERMAL ENGINEERING

I INTERNATIONAL JOURNALS

1. **Krishna, V. & Sharma, P.B. (n.d.).A model for combustion losses in a pulverized fuel fired power plant boiler. *Proc. I. Mech.E. (London), 202,* No. A4.**
2. **Krishna, V. & Sharma, P.B. (1990).A study of transients in turbine heat rate of steam power plant (UK).*Energy Conversion & Management, Boston, Texas, 2,* 187-202.**

3. **Krishna, V. & Sharma, P.B. (1984).Computer aided optimization of turbine heat rate of a thermal power plant, Trans. ASME. *Jr. of Energy Resources*.**

4. **Sharma, P.B. & Krishna, V. (1995). Improved formulation for turbine heat rate of a thermal power plant. *'Energy' International Journal*.**

5. **Pandey, R., Sethi, V.K.,Pandey, M., &Choubey, M. (2015). Recent research and various Techniques available for efficiency improvement of IGCC power plants. *Springer AISC Series/SCOUPS INDEXED JOURNAL,* "Artificial Intelligence and Evolutionary Computations in Engineering Systems", 49-58.**

6. **Pandey, R., Sethi, V.K., Pandey, M., &Choubey, M. (2015).Comparitivestudy of coal based IGCC and conventional power plants on various technological parameters. *IEEE Journal of Advances in Computing and Communication Engineering (ICACCE).***

7. **Pandey, R., Sethi, V.K., Pandey, M., &Choubey, M. (2016). Modelling and simulation of integrated gasification combined cycle (IGCC) power plant and its efficiency optimization.*Materials Today Elsevier*.**

8. **Pandey, R., Sethi, V.K., Pandey, M., &Choubey, M. (2017). A 100 MW integrated gasification combined cycle (IGCC) power plant: efficiency optimization. *Elsevier Journal of Advanced Research in Dynamical & Control Systems, 9*(5),212 – 220.**

9. **Pandey, R., Pandey, M., Sethi, V.K., &Choubey, M. (2017). Technology and resource management of IGCC power plants in the Indian context. *Elsevier Journal of Advanced Research in Dynamical & Control Systems, 9*(3),108-120.**

10. **Pandey, R., Pandey, M., Sethi, V.K., &Choubey, M. (2017). Potential analysis of Indian coal for power generation and pollution based on integrated gasification combined cycle (IGCC). *Elsevier Journal of Advanced Research in Dynamical & Control Systems,* 9(6), 212 - 220.**

II. NATIONAL JOURNALS

1. **Sharma, P.B., Sethi, V.K. "A Technique for computerized Thermal Power Plant Performance Monitoring", Jr. Irrigation and Power, Min. of Energy, India, pp. 417-428, July 1984.**
2. **Sethi, V.K. Sharma, P.B. and Gupta, SK "Power Plant Performance Analysis through Mathematical modeling", Jr. CEA, pp. 22-31, April 1983.**
3. **Sethi, V.K., Sharma, P.B. and Gupta, SK "Control Volume Approach for performance Testing of Boilers & Turbines", Jr. CEA, pp. 353-358, March 1985.**
4. **Sethi, V.K., Sharma, P.B. and Gupta, SK "Effects of Condenser Performance on Turbine Heat Rate of a Thermal Power Plant", Jr. of Thermal Engg. Vol. 4, No.2, pp. 46, 1985.**
5. **Krishna, Vinod and Sharma, P.B. "Some studies in Economic Aspects of Power Plant Performance". CBIP, Jr. Min. of Energy (India). Vol. 47, No.4, Oct.1990.**

III INTERNATIONAL CONFERENCES

1. **Krishna, V. and Sharma, P.B. "Uncertainty in Turbine Heat Rate Computation in an on-line Performance Monitoring System", Proc. 108th ASME Winter Annual Meeting, Dec. 1987, Boston, USA.**
2. **Sethi, V.K., Sharma, P.B. and Gupta, SK "A Mathematical Model for Turbine Heat Rate for a Steam Power Plant". Proc. Int. IEEE SMC Conf. pp. 1257-1262, Jan. 1984.**

3. Sethi, V.K., Sharma, P.B. and Gupta, SK "Modeling of Condenser Correction factor for Turbine Heat Rate for a Thermal Power Plant". Proc. Int. Joint Sym. of IS Chem. & AS Chem. Eng. Dec. 1985.

4. Sethi, V.K., Sharma, P.B. and Gupta, SK "A Model for Transients in Turbine Heat Rate for a Thermal power Plant". Proc. Int. Joint Sym. IS Chem. Engrs. and AS Chem. Eng. Dec. 1985.

5. Sharma, P.B. and Krishna, V. "Improvements in Mathematical Modeling of Turbine Heat Rate of a Thermal Power Plant". Proc. 23rd IECEC 1988 ASME, USA Aug. 1988.

VI. NATIONAL CONFERENCES

1. Sethi, V.K., Sharma, P.B. and Gupta, SK "Training in Power Plant Efficiency using a computer Aided Simulator". Proc. COMET-83, IIT, Delhi, March 1983.

2. Sethi, V.K., Gupta, SK and Sharma, P.B. "Analysis of Factors affecting the thermal performance of a power plant Boiler". Proc. Nat. HMT Conf. pp. 161-166, 1983.

3. Sethi, V.K., Sharma, P.B. and Gupta, SK "Turbine Heat Rate under Transient Conditions for a Thermal Power Plant". Proc. Nat. FMEP- Conf. pp. 127-133, Dec. 1983.

4. Krishna, V. et. al. "Computer Aided Optimization of Thermal Power Plant Performance". Proc. 13th Nat. FMEP Conf. Truchi (India), pp. 337-340, 1984.

5. Sharma R.K., Sethi V.K. "Performance Monitoring and Testing – Some Newer Techniques", Seminar and Energy optimization and Efficiency Improvement in Infrastructure & Industrial Sectors, IIPE, Delhi, Oct. 29-30, 1998.